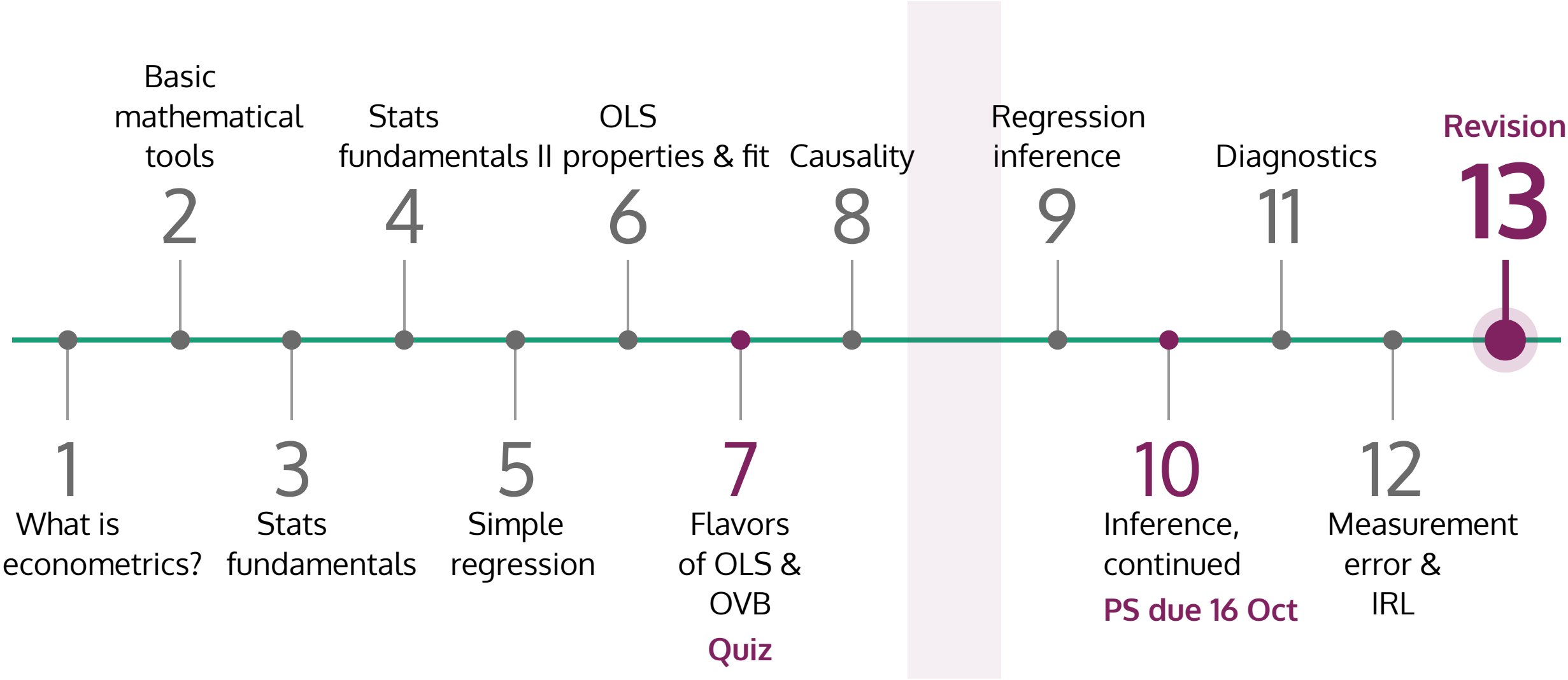


ECON 2041

Introductory Econometrics

Review

Plan for the semester



Plan for today

- Where students lost marks last year
- Reading a regression table
- Interactions
- Testing a claim with a confidence interval
- Omitted variables and heteroskedasticity
- Data that you can trust

Where students lost marks last year

Interactions: writing the model, interpreting the coefficient, testing it

Testing a value other than 0 with a confidence interval

The direction of omitted variable bias, once a proxy is added

Reading a regression table

Read each slope with the other xs fixed

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$$

β_1: average difference in y per 1 unit of x_1, holding x_2 fixed

Dummy x_2: average difference in y between groups 1 and 0, holding x_1 fixed

Airbnb prices: guests, rating, entire home

```
ols("price ~ guests + rating + entire_home", data=airbnb).fit(cov_type="HC3")
```

	coef	std err	z	P>|z|	[0.025	0.975]
Intercept	-269.6861	27.520	-9.800	0.000	-323.623	-215.749
guests	63.8244	1.357	47.023	0.000	61.164	66.485
rating	63.8766	6.005	10.638	0.000	52.108	75.646
entire_home	100.2121	4.231	23.686	0.000	91.920	108.504

`entire_home` = 1 for an entire home or apartment, 0 for a room

Data: Inside Airbnb, Sydney, snapshot of 16 June 2026; 12,419 listings

Open the portal on your phone

On the next slide: 4 sentences about the coefficient on `entire_home` in our Airbnb regression

Pick the one that's right

https://emiliatjernstrom.com/econ2041/portal.html



Reading the entire-home coefficient

$$\text{price} = \beta_0 + \beta_1\,\text{guests} + \beta_2\,\text{rating} + \beta_3\,\text{entire} + u$$

entire = 1 for an entire home, 0 for a room; $\hat{\beta}_3 = 100.21$

In all 4, guests and rating are held fixed

A. An entire home costs $100.21 a night, on average

B. An entire home costs $100.21 more a night than a room, on average ←

C. Turning a room into an entire home raises its price by $100.21 a night

D. An entire home costs 100.21% more a night than a room, on average

Vote on the portal: https://emiliatjernstrom.com/econ2041/portal.html

Log price: coefficient $\times 100 \approx$ % difference

```
ols("np.log(price) ~ guests + rating + entire_home",
    data=airbnb).fit(cov_type="HC3")
```

	coef	std err	z	P>|z|	[0.025	0.975]
Intercept	3.9228	0.059	66.401	0.000	3.807	4.039
guests	0.1299	0.002	59.573	0.000	0.126	0.134
rating	0.1585	0.013	12.125	0.000	0.133	0.184
entire_home	0.6012	0.015	39.382	0.000	0.571	0.631

1 more guest: about $100 \times 0.13 = 13\%$ higher price, holding rating and entire home fixed

Data: Inside Airbnb, Sydney, snapshot of 16 June 2026; 12,419 listings

Interactions

With an interaction, slopes differ by group

$$\text{price} = \beta_0 + \beta_1 \, \text{guests} + \beta_2 \, \text{entire} + \beta_3 \, (\text{guests} \times \text{entire}) + u$$

Rooms, entire = 0: slope on guests is β_1

Entire homes, entire = 1: slope on guests is $\beta_1 + \beta_3$

β_3: how much larger the slope is for entire homes than for rooms

Interactions in Python: a colon

```
ols("price ~ guests + entire_home + guests:entire_home",
    data=airbnb).fit(cov_type="HC3")
```

	coef	std err	z	P>|z|	[0.025	0.975]
Intercept	66.0203	12.544	5.263	0.000	41.435	90.605
guests	41.3797	6.030	6.862	0.000	29.561	53.199
entire_home	65.9176	13.653	4.828	0.000	39.157	92.678
guests:entire_home	23.0458	6.190	3.723	0.000	10.913	35.179

Data: Inside Airbnb, Sydney, snapshot of 16 June 2026; 12,419 listings

Reading the interaction coefficient

$$\text{price} = \beta_0 + \beta_1 \, \text{guests} + \beta_2 \, \text{entire} + \beta_3 \, (\text{guests} \times \text{entire}) + u$$

$$\hat{\beta}_3 = 23.05$$

A. For entire homes, the slope on guests is $23.05 per guest

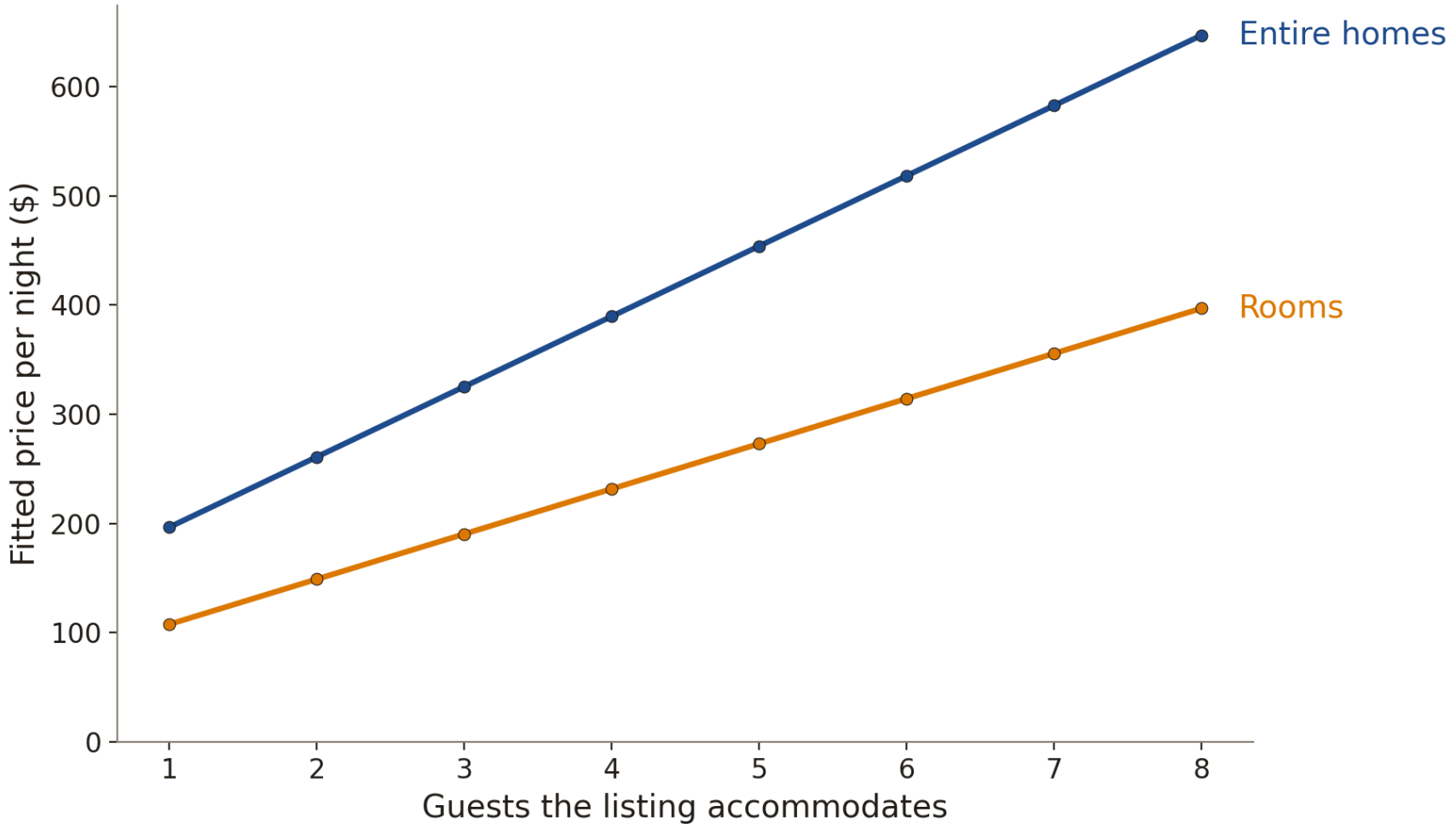
B. Entire homes have a slope on guests $23.05 steeper than rooms ←

C. Entire homes cost $23.05 more a night than rooms, holding guests fixed

D. Each extra guest is associated with a price $23.05 higher for every listing

Vote on the portal: https://emiliatjernstrom.com/econ2041/portal.html

Two slopes from one regression



Data: Inside Airbnb, Sydney, snapshot of 16 June 2026; 12,419 listings; fitted values for 1 to 8 guests

Is the slope different for entire homes?

$H_0 : \beta_3 = 0 \qquad H_A : \beta_3 \neq 0$

95% confidence interval for β_3: 10.91 to 35.18

0 is outside the confidence interval, so we reject H_0 at the 5% level

Testing a claim

Three tools, three questions

Question	Tool	Reject H_0 when
1 coefficient equal to a value?	t-test	$p < \alpha$
Plausible values of 1 coefficient?	Confidence interval	Value outside it
Several coefficients all 0?	F-test	$p < \alpha$

A host claims a $65 gap per extra guest

From our first table: coefficient on `guests` 63.82, 95% confidence interval 61.16 to 66.49

A. Reject: 65 isn't equal to the estimate, 63.82

B. Don't reject: 65 is inside the 95% confidence interval ←

C. Reject: the p-value for guests, 0.000, is below 0.05

D. Can't tell: the table's p-value is for 0 only

Vote on the portal: https://emiliatjernstrom.com/econ2041/portal.html

Test any value with a confidence interval

$65: inside 61.16 to 66.49, so we don't reject

$70: outside, so we reject at the 5% level

Not rejecting $65 doesn't show that $65 is true

Two regression problems

Two conditions for omitted variable bias

The omitted variable affects y, holding x fixed

The omitted variable is correlated with x

	Correlated + with x	Correlated − with x
Affects y +	Bias +	Bias −
Affects y −	Bias −	Bias +

Pokies: which way is the omitted variable bias?

Losses per adult on pokies per 1,000 adults: slope 78.3

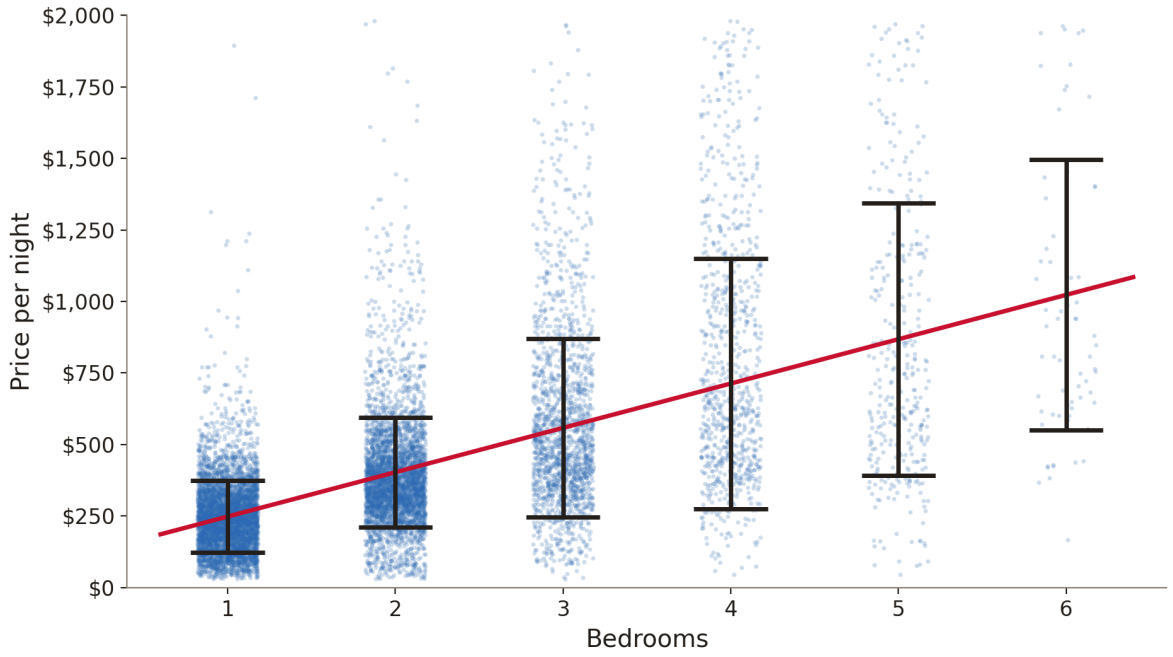
Unemployment affects losses: +

Unemployment is correlated with pokie density: +

Positive bias: the slope is 72.8 with unemployment in the regression

Data: VGCCC 2024, 79 Victorian LGAs

Heteroskedasticity: invalid standard errors



The error variance differs across observations

OLS is still unbiased

Usual standard errors are invalid: use robust ones, `cov_type="HC3"`

Data: Inside Airbnb, Sydney, snapshot of 16 June 2026; 12,419 listings

Data that you can trust

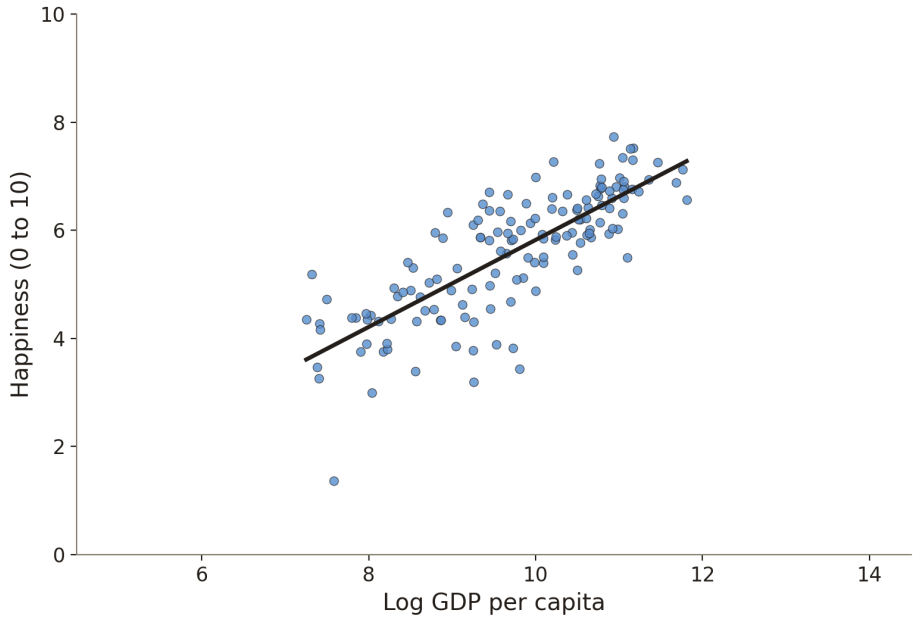
Three ways data go wrong

Structure: 1 row per observation, 1 column per variable

Quality: the values themselves are wrong, such as `-999` for missing, duplicates, mixed units

Measurement: the recorded values have error

With which noise is the slope flatter?



I'll add random noise to 1 variable

A. Only noise in happiness scores (y)

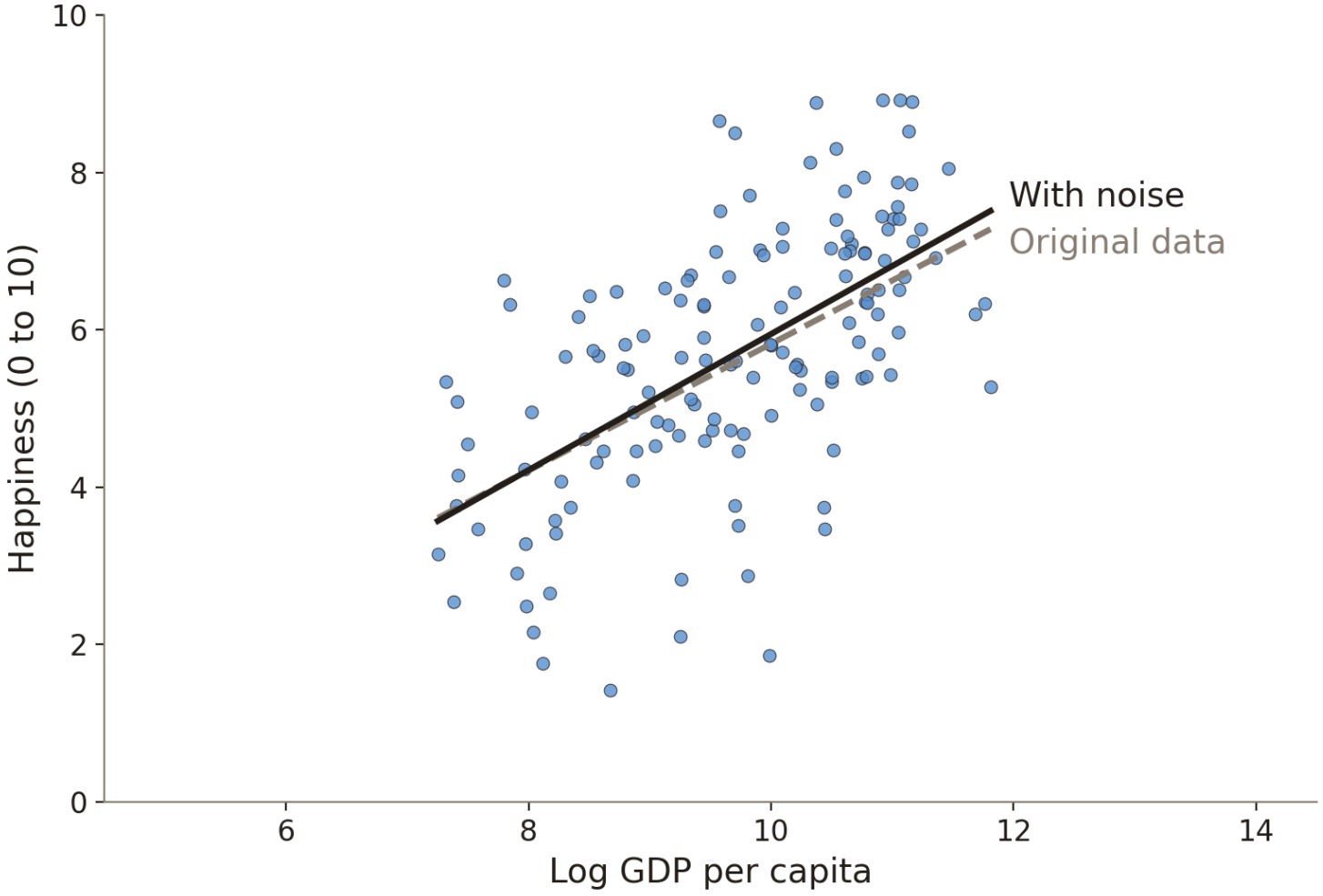
B. Only noise in log GDP (x) ←

C. Noise in either of the variables

D. Neither of the two kinds of noise

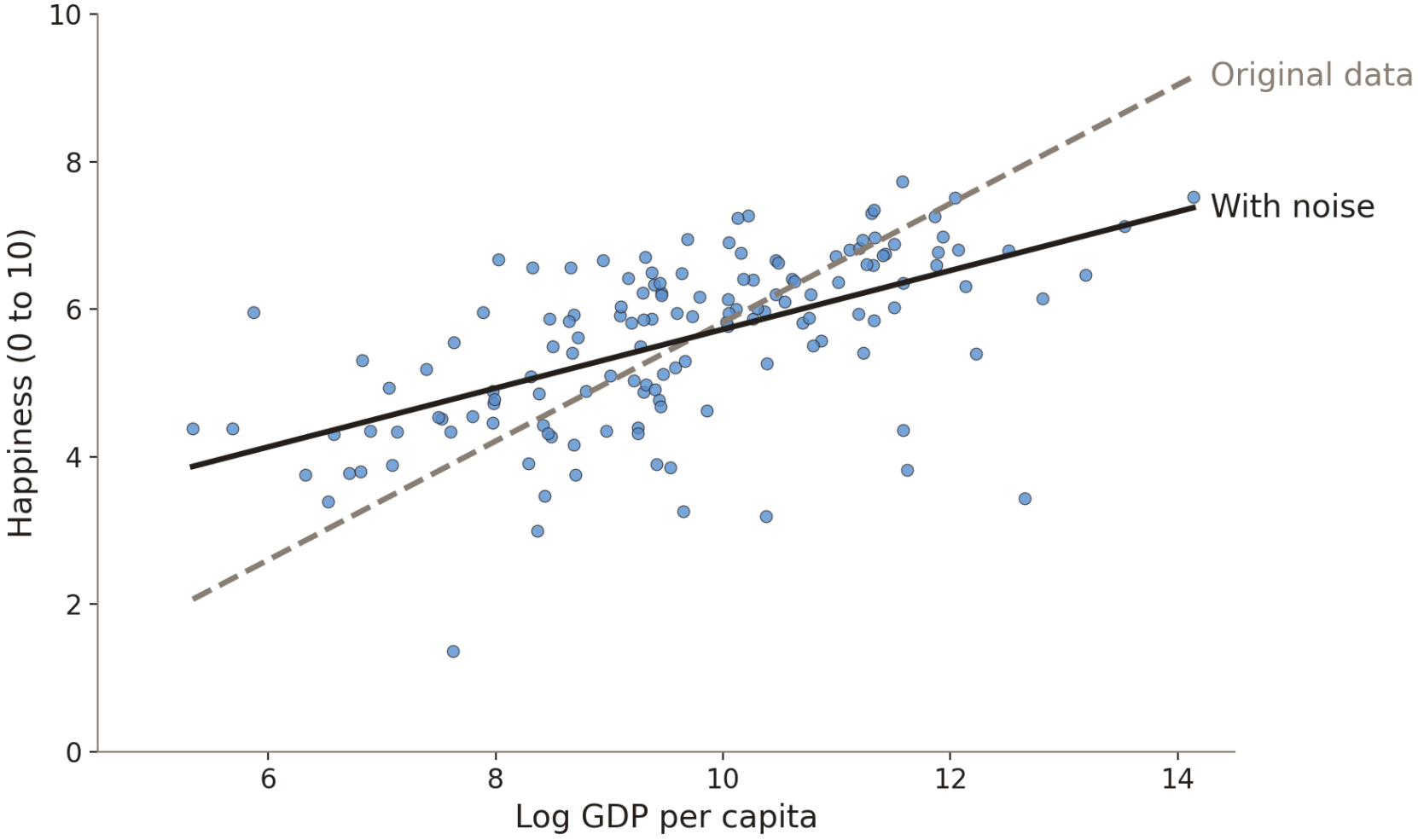
Data: Gallup World Poll & World Bank, via Our World in Data; 144 countries, 2024

Noise in y: similar slope, larger standard error



Data: Gallup World Poll & World Bank, via Our World in Data; 144 countries, 2024; noise added to happiness

Noise in x: a flatter slope



Data: Gallup World Poll & World Bank, via Our World in Data; 144 countries, 2024; noise added to log GDP

Good luck, and thanks!